
Diagnosing Representation Drift in Time-Series Foundation Model Fine-Tuning: A Moirai Case Study

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Abstract

We propose a **diagnostic protocol** for representation drift during fine-tuning of time-series foundation models. The protocol first tests whether zero-shot features are valuable relative to a linear baseline (value-gate), then uses controlled encoder comparisons — full fine-tuning against a frozen encoder — to ask whether encoder adaptation carries the downstream gain. In a Moirai case study spanning three model sizes and multiple datasets, fine-tuning often restructures representations toward the downstream objective, but the utility consequence depends on sample size and model capacity: Moirai-Small degrades in low-data ETTh2, while Moirai-Base and ILI fine-tuning improve despite substantial drift. Cross-backbone experiments (Chronos-T5-Small on M4-Monthly) confirm that the value-gate serves as a screening heuristic for when preservation-focused drift diagnosis is warranted. These results suggest that drift alone is insufficient evidence of harm; practitioners should first establish pre-training value, then assess whether drift affects task-relevant structure.

1 Introduction

Fine-tuning a time-series foundation model reshapes its internal representations. It is common knowledge that this reshaping moves features *toward* the downstream objective; the question we ask is different and, we argue, under-appreciated: **does the amount of representational change tell you whether fine-tuning helped or hurt?** Our central finding is that it does *not*. The same large drift degrades one model, improves another, and is irrelevant to a third; the outcome is governed by whether the pre-trained features were valuable to begin with and by model capacity, not by how much the encoder moved. We call this the *drift-utility dissociation*, and we contribute a diagnostic protocol that separates harmful forgetting from benign, task-directed specialisation.

Terminology. We write *zero-shot* (ZS) for the frozen pre-trained model evaluated without fine-tuning, and measure representational change with two diagnostics defined in §3: *centred kernel alignment* (CKA, geometric similarity between pre-trained and fine-tuned encoders; CKA=1 means no change) and *weight drift* (ℓ_2 parameter distance). A third, linear-probe ΔR^2 , appeared in an earlier version of this work and is **retracted** here (§6). We abbreviate learning rate as LR, sample size as n , and refer to the influenza-like-illness benchmark as ILI. Standard weight-anchoring baselines from continual learning—elastic weight consolidation (EWC Kirkpatrick et al. [2017]), L2-SP, and the parameter-efficient adapter LoRA—appear as mitigation conditions (§3).

Moirai Woo et al. [2024] is a foundation time-series model trained on billions of time points that achieves strong zero-shot performance on forecasting benchmarks. A natural next step is to fine-tune it on domain-specific tasks, but when should a practitioner worry about overwriting the representations that made zero-shot work? *Counterfactual:* Chronos-T5-Small on ETTh2 gives ZS

MSE=0.304 vs. a linear baseline of 0.213 (ZS *underperforms* by 43%). A practitioner who skipped screening would fine-tune Chronos, observe CKA dropping, and apply LoRA/EWC to preserve features that had *no advantage over Linear* (wasted effort). Our *value-gate* prevents this: preservation-focused drift diagnosis is most meaningful when pre-trained features demonstrably outperform a trivial baseline. Crucially, the gate is a *downstream-only* check (zero-shot evaluation vs. a linear regression on the practitioner’s own target data); it requires the released checkpoint and target set, never access to pre-training data or resources.

Representation drift during fine-tuning. We contribute a systematic *diagnostic methodology* for characterising representation drift (progressive overwriting of pre-trained features) during fine-tuning, and demonstrate it on Moirai (Small 13.8M, Base 91M, Large 311M)—the single backbone family whose features pass our value-gate on ETT. Three other backbones (Chronos Ansari et al. [2024], TimesFM-2.5 Das et al. [2024], MOMENT Goswami et al. [2024]) gate-fail on all nine ETT cells, the expected behaviour of a well-calibrated screen; Chronos gate-passes M4-Monthly (in its corpus) at 84.5%, confirming the gate identifies applicable cells across backbones (§5.1). Catastrophic forgetting during fine-tuning is well-studied in continual learning Kirkpatrick et al. [2017], Li and Hoiem [2018] and NLP Luo et al. [2023]; we use the term for degraded performance on distributions the model originally handled well (the “previous task” being the pre-training objective), and our contribution is reusable diagnostic methodology demonstrated on a controlled Moirai case study. Drift is distinguished from overfitting by the frozen-encoder control (CKA=1.000, same data/epochs): it isolates encoder weight updates as the cause, since freezing eliminates the degradation. Concretely, on ETTh2 (gate-pass, $R_{\text{task}}^2(\text{PT})=+0.279$) fine-tuning degrades MSE by 10–16% at $n=500$ and resolves by $n=2,000$; on IoT (sub-random ZS) drift is large but irrelevant—no valuable features existed to forget. When harmful, such drift degrades the distributional predictions used for anomaly scoring and probabilistic forecasting.

Scope. The empirical case study spans three Moirai gate-pass dissociation cells (ETTh2-Small, ETTh2-Base, ILI) plus a cross-backbone stability cell (Chronos-T5-Small/M4-Monthly, gate-pass 84.5%). The central finding is **non-monotone capacity dependence**: Base (91M) improves protocol-robustly at all n , Small (13.8M) degrades at low n , and Large (311M) degrades at default LR but is rescued by a $10\times$ LR reduction; a step-size falsification confirms capacity is an independent buffer against destructive drift (§5). We do not claim the dissociation is universal—the protocol applies to any gate-passing backbone but the numeric findings require broader replication. Two pre-registered predictions failed. The intermediate-CKA prediction for ILI is falsified outright (§5.1): its CKA component was wrong, and the probe component that had been reported as confirming it is withdrawn with the retracted axis (§6). The step-size hypothesis is partially falsified (Appendix G). Both are reported as such.

What is new here. That fine-tuning restructures representations toward the downstream objective is well established and is *not* our claim. Our contribution is that this restructuring is *dissociated* from task utility—large drift can be harmful, beneficial, or irrelevant—and a protocol that predicts *which* regime a given (backbone, dataset, capacity, n) cell is in *before* committing to a mitigation strategy.

Contributions.

1. **Diagnostic protocol (design is backbone-agnostic; empirically instantiated on Moirai).** Value-gate screening (ZS >20% vs. Linear; the threshold is a practical heuristic; gate-pass is reported as a continuous variable throughout and we do not claim universality of this cut-off), and controlled-comparison design (frozen-encoder, random-init, layer-wise unfreeze). The protocol applies to any backbone; the drift–utility dissociation it reveals is established here on the Moirai family, with Chronos/M4 confirming the value-gate screen on a second backbone (§5.1). A reproducibility bundle (configs, seed lists, metric scripts, uni2ts patch, expected outputs) accompanies this paper.
2. **Empirical validation: dissociation and stability regimes.** *Dissociation:* on Moirai/ETTh2 the encoder drifts far below the 0.95 flag while forecasting improves, and across the sample-size sweep drift moves monotonically while the task outcome does not. *Stability control:* Chronos/M4 (gate-pass 84.5%, $\text{CKA}\geq 0.95$; §5.1): no restructuring, and the frozen encoder matches full fine-tuning.

2 Background and Related Work

Catastrophic forgetting in fine-tuning. Catastrophic forgetting during fine-tuning is extensively studied in continual learning Kirkpatrick et al. [2017], Li and Hoiem [2018] and NLP Luo et al. [2023]. Kumar et al. [2022] show full fine-tuning can distort pre-trained features relative to linear probing; Andreassen et al. [2023] characterize OOD robustness decay throughout fine-tuning. Our contribution is a systematic *characterization* for time-series foundation models: CKA, linear probing, weight drift, and frozen-encoder controls across domains with contrasting pre-training value.

Time-series foundation model adaptation. Recent work on adapting pre-trained time-series models includes probabilistic fine-tuning (Lag-Llama Rasul et al. [2024]), generative pre-training with task-specific heads (Timer Liu et al. [2024]), LLM reprogramming (Time-LLM Jin et al. [2024]), and cross-task transfer (One Fits All Zhou et al. [2023]). These works develop *adaptation recipes*; our contribution is orthogonal: a *diagnostic methodology* for characterising what happens to pre-trained representations during any such adaptation, applicable when the practitioner needs to decide whether drift is harmful or benign.

Analysing time-series foundation model representations. A parallel line of work probes the internal structure of TSFMs directly. Wiliński et al. [2025] characterise and *steer* representations of *frozen* TSFMs, uncovering block-like redundancy and interpretable concepts (periodicity, trend) that can be added by latent-space intervention. Our study is complementary along the time axis: rather than intervening on a fixed model, we ask how those representations *change under fine-tuning* and whether the change is harmful. Our diagnostics build on representation-similarity (CKA Kornblith et al. [2019]) and probing methodology Hewitt and Manning [2019], Pimentel et al. [2020], applied here to the pre-trained-vs-fine-tuned encoder pair rather than to a single frozen model.

Setting. Given a pre-trained time-series foundation model with encoder $f_\theta : \mathbb{R}^{T \times F} \rightarrow \mathbb{R}^d$ and downstream dataset $\mathcal{D}$ of size n , fine-tuning produces updated parameters θ' by minimising a task loss (NLL for forecasting). We measure encoder changes using two diagnostics (§3): (1) CKA between f_θ and $f_{\theta'}$ (geometric similarity) and (2) weight drift $\|\theta' - \theta\|_2$. A linear-probe ΔR^2 diagnostic used in an earlier version is retracted (§6).

Pre-training value gate. Studying whether fine-tuning destroys valuable features requires a domain where such features are demonstrably present. We set the gate at zero-shot MSE 20% below a Linear baseline; ETTh2 at $h=96$ passes with Moirai-Small ($R^2_{\text{task}}(\text{PT})=+0.279$). The gate is deliberately cheap and *downstream-only*: the practitioner—defined here as someone adapting a *released* foundation-model checkpoint to a small target dataset—runs the frozen model zero-shot on the target set and compares to a lookback-96 linear regression fit on that same set. Neither pre-training data nor pre-training compute is required, so the screen is available exactly in the small-sample adaptation regime where fine-tuning is normally performed. ETTh1 and ETTm2 gate-fail at the standard evaluation protocol ($R^2_{\text{task}}(\text{PT})=-0.41$ and -0.56 respectively; Table 4); they are included as cross-domain comparisons to characterise drift behaviour outside the gate-pass regime. Domains that gate-fail serve as negative controls. The 20% threshold is calibrated to ETT short-horizon benchmarks where linear baselines are surprisingly competitive Zeng et al. [2023]: a model must substantially outperform a lookback-96 linear regression before drift can be *harmful* (rather than irrelevant). The threshold is a practical screen, not a universal constant; we report $R^2_{\text{task}}(\text{PT})$ values for each cell (Table 4) so readers can apply alternative thresholds.

Task-level consequence: forgetting. We define forgetting as signed relative change in zero-shot-canonical task metric on held-out test: $\text{forg.\%} = 100 \cdot (M_{\text{FT}} - M_{\text{ZS}}) / |M_{\text{ZS}}|$ (MSE for forecasting, so positive = worse). $\text{Forg.} > 0$ indicates catastrophic forgetting (degraded performance vs. pre-training baseline).

Decoupling harmful forgetting from benign specialisation. A reviewer-critical distinction: representational change that redirects features toward the target (*benign specialisation*) must be separated from change that destroys generally useful structure (*harmful forgetting*). An earlier version of this work operationalised the distinction with a *probe-sign asymmetry* criterion—trained-objective decodability up, task-orthogonal decodability flat or down. That criterion is retracted (§6): the asymmetry appears only where the probes fail to generalise and vanishes once they do, so it cannot separate benign specialisation from harmful forgetting. What remains is *value-gate conditioning*.

Representational change is harmful only when the structure it discards was valuable, i.e. in a gate-pass cell ($R_{\text{task}}^2(\text{PT}) > 0$, ZS beats Linear); in a gate-fail cell the same change is benign because no valuable features existed to forget. Thus forgetting > 0 in a gate-pass cell is the operational definition of harm, and the question of whether the encoder’s change *carried* the gain is answered by intervention—comparing full fine-tuning against a frozen encoder—not by observing the representation. This is the mechanism the diagnostics in §3 (notation in Table 1) are designed to resolve.

3 Fine-Tuning Protocol

Moirai for forecasting and anomaly scoring. Moirai Woo et al. [2024] predicts a probabilistic distribution over a target horizon given context. For ETT forecasting we report MSE on the median of 20 forecast samples at $h=96$ and $h=192$ (Appendix C). For IoT the anomaly score is the fraction of timesteps outside the $1-\alpha$ CI ($\alpha=0.05$) over a 32-timestep horizon.

Fine-tuning objective. Condition B uses NLL only. Condition C adds supervised contrastive loss Khosla et al. [2020]: $\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y}; \tau)$, with $\lambda = 0.5$ and $\tau = 0.07$. For ETT, temporal clusters (8 bins by time-series position) serve as SupCon label.

Training details. AdamW ($\eta = 10^{-4}$, weight decay 10^{-2}), batch size 32, early stopping with patience 3, gradient clipping at 1.0. ETT fine-tuning runs 20 epochs at $n=500$ (10 seeds) and is swept across $n \in \{500, 1k, 2k, 5k, 10k\}$. All Moirai sizes (Small 13.8M, Base 91M, Large 311M) use the same protocol.

Deterministic training protocol. All $n=10k$ per-seed results use CUDA-deterministic training: `CUBLAS_WORKSPACE_CONFIG=:4096:8`, `torch.backends.cudnn.deterministic=True`, seeded DataLoader workers, and fixed PYTHONHASHSEED. This makes the 10-seed per-seed comparison reproducible: any researcher can recover identical per-seed forgetting values by re-running with the same seeds on the same GPU architecture (A10G). Full seed list, per-seed values, and hardware specification are in Appendix I. Cross-hardware reproducibility (e.g., A100 vs. A10G) has not been empirically verified; the deterministic guarantee applies within the same GPU architecture (A10G, as used in this paper).

uni2ts PackedStdScaler patch.¹ Any Moirai fine-tuning reproduction of this paper requires the patch.

Representation diagnostics. We track representation change with three metrics: (1) CKA Kornblith et al. [2019] between pre-trained and fine-tuned encoder representations (geometric similarity); and (2) weight drift (ℓ_2 distance from pre-trained parameters). An earlier version added linear-probe ΔR^2 as a third diagnostic; because its absolute $R^2(\text{FT})$ is deeply negative, it measures relative change in a regime where the probe does not generalise, and we retract it (§6).

Table 1: Notation summary for probe-related quantities.

Symbol	Definition
$R_{\text{task}}^2(\text{PT})$	$1 - \text{MSE}_{\text{ZS}}/\text{MSE}_{\text{Linear}}$; positive $\Rightarrow$ gate-pass
$R^2(\text{PT}), R^2(\text{FT})$	Ridge probe score on pre-trained / fine-tuned encoder (retracted axis)
ΔR^2	$R^2(\text{FT}) - R^2(\text{PT})$; retracted , see §6
ΔR_{task}^2	Task-level improvement measured via forecasting head

Rationale for experimental conditions. Each condition (used throughout §4–§5) answers one question, spanning the axis from unconstrained fine-tuning (maximum drift) to a frozen encoder (zero drift); Table 2 makes the mapping explicit. This lets us attribute any observed effect to a specific intervention rather than to fine-tuning in the aggregate.

¹We patched an in-place tensor mutation in uni2ts’s PackedStdScaler that silently detaches autograd; a differentiable `torch.where` replaces the in-place write. All results use the patched scaler (released with our code). See Appendix B for details.

Table 2: Experimental conditions and the question each isolates.

Cond.	Setting	Question it isolates
B	NLL-only fine-tune	Baseline: how much does unconstrained fine-tuning drift/forget?
C	NLL + SupCon	Does an explicit representation-shaping loss prevent drift?
D	Frozen encoder	Control: separates encoder updates from head training / overfitting
E	LoRA ($r=8$)	Can low-rank adaptation preserve the encoder while adapting?
F	L2-SP	Standard weight-anchoring baseline (penalise deviation from PT)
G	EWC	Fisher-weighted anchoring baseline from continual learning

4 Representation Drift Diagnosis: ETT Forecasting

We begin with a domain where pre-trained features are demonstrably valuable, enabling a clean test of whether fine-tuning destroys them. ETT time-series forecasting benchmarks Zhou et al. [2021] provide this setting: zero-shot Moirai outperforms Linear baselines by 28–45% (Section 4.1). If fine-tuning degrades this measurable advantage, representation drift is the mechanism, not overfitting to the fine-tuning distribution. The frozen-encoder control (same data, same epochs, CKA=1.000) confirms this: overfitting would not be eliminated by freezing.

4.1 Experimental Setup and Value Gate

We evaluate on ETTh2 (7 features, 17k timesteps, 12/4/4 month split). Moirai-Small (13.8M) zero-shot outperforms Linear by 28–45% (ZS MSE: 0.126 at $h=96$, 0.158 at $h=192$; un-normalised gate-pass scale: 0.269 vs. 0.373 at $h=96$, 0.299 vs. 0.544 at $h=192$), passing the value gate (20% threshold). The 20% threshold is a practical screen, not a phase boundary: at 10–20%, drift diagnosis applies but marginal features reduce both harm from drift and benefit from mitigation. Conditions: **B** (NLL-only), **C** (NLL+SupCon), **D** (frozen encoder), **E** (LoRA $r=8$), **F** (L2-SP), **G** (EWC). Training: $n=500$ samples, 20 epochs, AdamW (lr=1e-4), 10 seeds (B/C/D) or 5 seeds (E/F/G). Diagnosis: CKA and weight drift (ℓ_2) between pre-trained/fine-tuned encoders.

4.2 Representation Drift Degrades High-Value Features

Fine-tuning degrades ETTh2 performance where pre-trained features are valuable (Appendix Table 6). Conditions B (NLL-only) and C (NLL+SupCon) degrade by 10–16% MSE (10/10 seeds positive, $p=0.002$ sign test; $d=1.23$, 95% CI [0.37, 2.01]; bootstrap $p<10^{-4}$) while training loss *decreases*, not overfitting (which raises validation loss), but representation drift. The frozen-encoder control (D, CKA=1.000, same data/epochs) eliminates drift, ruling out the training regime. Degradation scales with horizon: worse at $h=192$ (13–16%) than $h=96$ (10–11%). Contrastive loss (C) does not prevent drift: C (+10.1%) and B (+11.1%) differ by 1.0%, within seed standard deviation; a null result for the contrastive objective.

4.3 Mitigation Spectrum

We evaluate five mitigation strategies spanning unconstrained fine-tuning (most drift) to full freezing (Appendix Table 7).

LoRA achieves the best trade-off. LoRA ($r=8$, $\alpha=16$) *improves* over zero-shot at $h=96$ while maintaining high CKA (0.979) and zero weight drift. On Moirai-Large the same recipe fails at default LR (+25.7%, rank escalation does not rescue) but a $10\times$ LR reduction does (forg.= −8.5%, CKA=0.989), transferring to ETTh1/ETTh2 at $n=500$: **tune LR before rank** across model sizes. By contrast, EWC and L2-SP reduce but do not eliminate drift, and buy no reliable utility: matched at $n=1000$ and $h=96$, each measured against its own zero-shot, they land at +13.4 to +16.0% above ZS against full fine-tuning’s +13.4%—one worse, one a dead heat at markedly higher CKA (.965 vs .936), so anchoring is *not* uniformly harmful, merely uninformative. (An earlier version quoted 18–19% here, from a table that divides by a single shared zero-shot and mixes $n=500$ and $n=1000$ rows; see the caption of Table 7.) The frozen encoder eliminates drift (CKA=1.000) but cannot adapt when pre-trained features are insufficient.

Table 3: Sample-size sweep on ETTh2 ($h=96$, condition B). Drift grows monotonically (CKA $\downarrow$ as $n\uparrow$); forgetting is *non-monotonic*: severe at $n\leq 1,000$, resolved at $n=2,000$, with elevated variance at $n=5,000$ (crossover regime). At $n=10,000$: CUDA 10-seed deterministic early-stopped, 7/10 negative forgetting. **The two ΔR^2 columns are retracted** (§6) and are reproduced only as a record of the withdrawn computation: absolute R^2 (FT) is negative throughout ($R^2_{\text{ZS}}=-6.90$), i.e. the probe never decodes the target, and the effect vanishes once the probe does. Read the MSE, Forg.% and CKA columns only.

Samples	MSE	Forg.%	CKA	R^2 (FT)	ΔR^2 (Ridge)	ΔR^2 (MLP)
500	.148 $\pm$.008	+14.0 $\pm$ 6.7	.949 $\pm$.010	-6.79 ($k=2$)	+.12 ($k=2$)	-.26 ($k=1$)
1,000	.147 $\pm$.010	+13.4 $\pm$ 9.1	.936 $\pm$.025	-6.77 ($k=1$)	+.13 ($k=1$)	+.30 ($k=1$)
2,000	.126 $\pm$.008	-2.9 $\pm$ 5.9	.814 $\pm$.042	-6.51 ($k=3$)	+.39 $\pm$.35	+.73 $\pm$.75
5,000	.134 $\pm$.012	+6.6 $\pm$ 10.0	.546 $\pm$.082	-6.41 ($k=7$)	+.50 $\pm$.26	—
10,000 ^{cuda}	.122 $\pm$.008	-5.3 $\pm$ 6.2	.518 $\pm$.188	—	+.67 $\pm$.23 [†]	—

^{cuda} CUDA 10-seed deterministic (A10G); all completed, mean best epoch 4.2 $\pm$ 1.4 (Appendix I for per-seed values and protocol comparison). [†]Retracted (§6). Forgetting sign: 7/10 negative (95% Wilson CI [0.40, 0.89]; protocol-dependent—see §5).

Why run the diagnostic if LoRA is the best mitigation? Because the diagnostic decides *whether preservation is worth doing at all*, and reveals when LoRA itself fails. (i) On gate-fail cells (IoT, ETTm2) there is nothing to preserve; full fine-tuning is fine and applying LoRA/EWC to protect worthless features is wasted effort—the value-gate is what tells the practitioner to skip preservation entirely. (ii) LoRA is *not* uniformly safe: LoRA-Large fails at the default LR (+25.7%) and only the capacity/step-size diagnosis surfaced the 10 $\times$ LR fix. So the diagnostic and the mitigation are not competitors—the diagnostic is the *decision procedure* (preserve? which regime? which LR?) and LoRA is the recommended *action* once the diagnostic says preservation is warranted.

4.4 Sample-Size Sweep: Drift–Forgetting Dynamics

Table 3 ($n\in\{500, \dots, 10k\}$, ETTh2) reveals that *drift and forgetting are dissociable*, a finding best illustrated by comparing Moirai-Base (protocol-robust improvement of 50–57%, 10/10 seeds, despite CKA $\approx$ 0.25; §5) with Small, where the same mechanism produces capacity-dependent outcomes. On Small, drift grows monotonically with n (CKA from 0.949 at 500 to 0.518 at 10k), but forgetting is non-monotonic: severe at low n (+14% at 500), resolved by 2k (-2.9%), predominantly negative at 10k (-5.3 $\pm$ 6.2%, 7/10 negative, 10-seed CUDA deterministic). The $n=5k$ variance (+6.6 $\pm$ 10.0%, 7 seeds) is bimodal: 4 seeds cluster at -0.1%, 3 at +15.5% with epoch-1 overshoot (full trajectory analysis in Appendix I.1). Drift resolution at high n is protocol-dependent for Small (final-epoch -0.6%, n.s.) but protocol-robust for Base (§5).

Headline: drift is capacity-dependent: harmful on Small at $n\leq 1k$ (where practitioners most need pre-trained features), resolved by larger capacity (Base improves at all n , protocol-robust).

The insight in Table 3 is not any single row but the *divergence of two columns*: CKA falls monotonically as n grows (drift strictly increases), yet forgetting is non-monotonic and reverses sign (harm at low n , resolution at high n). If drift magnitude predicted harm, these columns would track; they do not. This is the drift–utility dissociation in a single table, and it is why a practitioner cannot read task risk off a CKA number—the decision needs the value-gate and the sample-size regime, not the drift magnitude alone.

The joint trajectory is plotted in Fig. 1; panels (a) and (d) carry the claim—a monotone drift curve beside a non-monotone outcome curve.

Task-native head: the task score that survives. Using Moirai’s own forecasting head gives a well-defined positive-floor task score, $R^2_{\text{task}}(\text{PT}) = 1 - \text{MSE}_{\text{ZS}}/\text{MSE}_{\text{Linear}} = 1 - 0.269/0.373 = +0.279$ at $h=96$ (+0.450 at $h=192$). After fine-tuning ($n=10k$, ES) it improves to $\text{MSE}_{\text{FT}}=0.122\pm 0.006$ vs. 0.126 ZS ($\Delta R^2_{\text{task}} > 0$ in 7/10 seeds; Table 4): the geometry shift is functionally beneficial at high n , harmful at low n (forgetting +14% at $n=500$). This quantity is a forecast-MSE ratio, not a linear-probe score, and is unaffected by the retraction below.

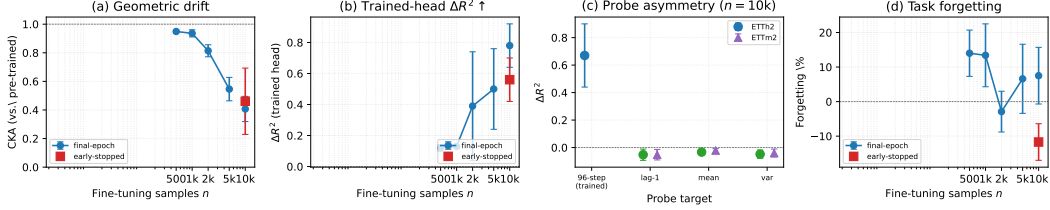


Figure 1: Cost-benefit dynamics on Moirai-Small/ETTh2 (condition B, $h=96$). (a) CKA decays monotonically with n (final-epoch vs. early-stopped markers); (d) forgetting is non-monotonic and protocol-dependent. Panels (b) and (c) plot the linear-probe ΔR^2 statistic and its sign asymmetry, both **retracted** (§6); they are reproduced unaltered as a record of the withdrawn analysis and support no claim here. The dissociation this figure is cited for is the contrast between (a) and (d): drift monotone in n , outcome not.

Table 4: Task-native score R^2_{task} at key (n, h) operating points. **Not a linear probe:** this is a ratio of forecast MSEs computed through Moirai’s own forecasting head, and is unaffected by the retraction of the linear-probe ΔR^2 axis (§6). (condition B, mean over available seeds). $R^2_{\text{task}}(\text{PT}) = 1 - \text{MSE}_{\text{ZS}} / \text{MSE}_{\text{Linear}}$ uses the gate-pass evaluation scale (median Moirai prediction, lookback=192, linear baseline lookback=96, train-normalised). $\Delta R^2_{\text{task}} = -(1 - R^2_{\text{task}}(\text{PT})) \times \text{Forg}\% / 100$ (sign is scale-independent; see §4.4). The $n=10\text{k}$ rows use early-stopped (ES) forgetting. ETTh2 is the **gate-pass** anchor ($R^2_{\text{task}}(\text{PT}) > 0.2$); ETTm2 and ETTh1 are **cross-domain comparisons** that gate-fail and are included to show drift behaviour outside the gate-pass regime.

n	h	$R^2_{\text{task}}(\text{PT})$	Forg.%	ΔR^2_{task}	$R^2_{\text{task}}(\text{FT})$
<i>ETTh2-Small (gate-pass: $R^2_{\text{task}}(\text{PT}) > 0.2$, Moirai outperforms Linear by 28%)</i>					
500	96	+0.279	+11.1	−0.080	+0.199
500	192	+0.450	+15.7	−0.086	+0.364
10k (ES)	96	+0.279	−5.3	+0.038	+0.317
<i>ETTh2-Base (gate-pass: $R^2_{\text{task}}(\text{PT}) = +0.31$, Moirai-Base outperforms Linear by 31%)</i>					
500	96	+0.310	−56.9	+0.393	+0.703
10k (ES)	96	+0.310	−54.2	+0.374	+0.684
<i>ETTh2 (gate-fail: $R^2_{\text{task}}(\text{PT}) < 0$, Moirai worse than Linear by 55%)</i>					
500	96	−0.556	−13.3	+0.207	−0.349
10k (ES)	96	−0.556	−28.2 $\pm$ 2.2	+0.439	−0.117
<i>ETTh1 (gate-fail: $R^2_{\text{task}}(\text{PT}) < 0$, Moirai worse than Linear by 41%)</i>					
500	96	−0.406	+7.0	−0.098	−0.505
10k (ES)	96	−0.406	−4.6 $\pm$ 4.1	+0.065	−0.342

Reading: ETTh2 (gate-pass): at $n=500$, $\Delta R^2_{\text{task}} < 0$ (forgetting); at $n=10\text{k}$ ES, $\Delta R^2_{\text{task}} > 0$ (improvement) in 7/10 seeds. ETTm2 and ETTh1 (gate-fail): fine-tuning can still improve over the ZS baseline ($\Delta R^2_{\text{task}} > 0$) even when the ZS baseline itself is weaker than linear regression ($R^2_{\text{task}}(\text{PT}) < 0$). The ΔR^2_{task} sign equals the sign of improvement over ZS baseline regardless of scale or gate status. ETTm2 seeds: $k=5$ (v19 CUDA); ETTh1 seeds: $k=10$ (v21 CUDA), 9/10 negative at $n=10\text{k}$ ES.

Retracted: linear-probe analyses of this cell. An earlier version continued here with a linear-probe account of *what* restructured—a task-orthogonal probe battery, a horizon sweep, and a layer-wise localisation claim separating geometric reorientation (top layers) from task recovery (lower layers). All of it rested on ΔR^2 and is retracted (§6); the underlying numbers are retained in Appendices I, K.1 and N as a record of the withdrawn computation. The layer-wise *task* readings do survive, since they are forecast MSE: unfreezing the top 3 of 6 layers and unfreezing all of them both reach forgetting $\approx -6.5\%$, at CKA 0.663 and 0.387 respectively—so matching the task outcome does not require matching the drift.

Table 5: Diagnostic summary. Gate=gate-pass%; Drift=CKA<0.95. The Asym. column recorded the probe-sign asymmetry, which is **retracted** (§6); it is retained only as a record of the withdrawn computation and carries no claim. All Moirai-Small unless noted. Full numerics in text.

Setting	Gate%	Drift?	Asym.?	Effect	Interpretation
ETTh2-S low- n	28	mild	10/10	Degrades	Restructuring harms (data-poor)
ETTh2-S high- n	28	yes	10/10	Mixed*	Resolves with sufficient n
ETTh2-B (Base)	31	strong	9/10	Improves	Capacity buffers drift
ILI	57	yes	10/10	Improves	Restructuring toward task
Chronos/M4	84.5	no	—	Stable	Encoder preserved (strong gate)
ETTh2 (gate-fail)	<0	yes	10/10	Improves	No features to preserve
IoT (gate-fail)	<0	strong	—	Improves	Replacement, not forgetting

*Protocol-dependent: -5.3% early-stopped, $+7.5\%$ final-epoch. Gate-pass = $1 - \text{MSE}_{\text{ZS}} / \text{MSE}_{\text{Linear}}$.

5 Cross-Domain Analysis and Discussion

Task forgetting is protocol-dependent. Forgetting is non-monotonic in n ($+14\%$ at 500, -2.9% at 2k, $-5.3 \pm 6.2\%$ at 10k ES, 7/10 negative) with protocol-dependent sign (final-epoch $+7.5 \pm 8.2\%$): of 10 CUDA seeds at $n=10\text{k}$, 7 converge monotonically (best epochs 3–5) and 3 overshoot late (epochs 12–18), the source of the sign difference (Appendix I.1). We therefore report the sign of forgetting at this cell as protocol-dependent rather than settled, and do not lead on it.

Second cell: ETTm2 $n=10\text{k}$ drifts with the *opposite* task outcome. ETTm2 ($h=96$, $n=10\text{k}$, CUDA $k=10$) drifts substantially yet forgetting is $-29.2 \pm 2.1\%$ —a large improvement—in a cell that *fails* the gate. Drift there overwrites features that were not outperforming Linear, so the same magnitude of representational change is benign. Together with ETTh2, this is the dissociation in its clearest form: comparable drift, opposite consequence, and the gate—not the drift reading—is what separates them.

ETTh1 $n=10\text{k}$: boundary case. On ETTh1 ($h=96$, $n=10\text{k}$, $k=10$), forgetting resolves ($-6.1 \pm 3.6\%$) while CKA falls to 0.727: another cell where substantial drift accompanies improvement rather than harm.

Cross-domain envelope. Table 5 summarises all settings tested; drift occurs throughout but consequences depend on pre-training value.

Capacity sensitivity: Moirai-Base as second gate-passing cell. Moirai-Base (91M) gate-passes ETTh2 ($R^2_{\text{task}}(\text{PT})=+0.31$) but behaves qualitatively differently from Small: task *improvement* at both n (forg. -56.9% at 500, -54.2% at 10k; 10/10 negative) despite *aggressive* drift (CKA 0.25–0.33): the most heavily drifted cell in the study is also one that improves. Base is protocol-robust and improves even at low n where Small degrades. Moirai-Large (311M) mirrors Small at $n=500$ ($+10.5\%$); LoRA-Large is rescued by $10\times$ LR reduction (Appendix F). **Step-size falsification:** Base at $10\times$ LR restructures more aggressively than Small at default LR (CKA=0.135 vs. 0.937) yet *still improves* (-58.2% , 10/10 positive), while Small at $0.1\times$ LR merely near-freezes (CKA=0.987). Thus capacity is an independent buffer against destructive drift, not merely a lower effective step size (Appendix G).

IoT negative control. Moirai ZS AUC=0.481 (sub-random); drift (CKA 0.31–0.45) is benign because from-scratch CNN matches fine-tuned Moirai; the value-gate correctly screens out this cell (Appendix K).

5.1 Non-Moirai Backbones: All Nine Cells Gate-Fail

Matched-protocol ZS across nine non-Moirai backbone $\times$ ETT cells (Chronos, TimesFM-2.5, MO-MENT $\times$ ETT) all gate-fail (-14% to -148% vs. Linear; Appendix A). Corpus composition is the most parsimonious account: **Chronos-T5-Small gate-passes on M4-Monthly** (84.5%, in its training corpus) while gate-failing on ETT (-43%)—same architecture, opposite gate outcome; and Moirai-Small itself gate-fails on Traffic/Exchange/ Electricity, so *pre-training emphasis* rather than mere corpus inclusion determines gate-pass. On M4-Monthly, full diagnostics (10 seeds, conditions

B/D) give $\text{CKA} \approx 0.98\text{--}0.999$ and no distinguishable forgetting ($+0.5\%$ at $n=10k$): strong gate-pass coincides with encoder stability, consistent with the value-gate as a regime indicator (Appendix O).

ILI: third gate-passing cell (10 seeds). Moirai-Small on ILI (weekly influenza, $h=24$, gate-pass=57%, conditions B/D) yields $\text{CKA}=0.478 \pm 0.022$ (restructuring) with large task improvement (forgetting= -80.7% , 10/10 negative). **Correction.** We previously reported that the frozen-encoder control (D) gives *zero* improvement in all 10 seeds, and concluded that the encoder is the sole locus of adaptation ($B-D=-80.7$ pp). That is withdrawn. The script short-circuited condition D: it assigned the zero-shot MSE as the fine-tuned MSE and wrote a forgetting of exactly 0.0, so no frozen-encoder model was ever trained and the “0.0% in all 10 seeds” was a single constant, not a measurement. Re-running condition D so that it trains — encoder frozen, the remaining parameters updated, as in the Moirai cells of §4 — gives $B-D=-62.6 \pm 4.6$ (10/10 seeds), not -80.7 : the frozen encoder recovers a large share of the gain rather than none of it. Its CKA is 0.76–0.90, not the asserted 1.000, because the projections feeding the encoder remain trainable. Condition B is unaffected and reproduces (-80.0 ± 2.0 against the published -80.7 ± 0.9 ; CKA 0.456 against 0.478), so the entire discrepancy was the untrained control. Per-seed values accompany the released code. The pre-registered CKA-intermediate prediction (0.90–0.95) is **falsified**—actual CKA is 0.478. This revises the spectrum hypothesis: restructuring extends from 28% through 57% gate-pass, with stability beginning above $\approx 80\%$ (Chronos/M4).

6 Retraction: The Probe-Sign Asymmetry Axis

An earlier version of this work reported a second diagnostic axis alongside CKA drift: a *probe-sign asymmetry*, in which fine-tuning raised linear decodability of the trained objective ($\Delta R^2 > 0$) while leaving task-orthogonal probes flat or lower. We reported it as replicating across every gate-passing cell. **We retract it.** It does not survive a protocol with every choice fixed in advance, and nothing in this paper’s remaining claims depends on it.

What fails. The statistic reproduces numerically—the headline $\Delta R^2 = +0.67 \pm 0.23$ on Moirai-Small/ETTh2 at $n=10k$ is recoverable to four significant figures—but only in a regime where the probe does not generalise. Under a temporal split the pre-trained probe scores $R^2(\text{PT})$ between -2.7 and -164.1 : it is not decoding the target at all, so ΔR^2 measures relative movement between two failed fits. Re-running the identical probes, encoders and windows under a randomised split, where all seven probes generalise ($R^2(\text{PT}) \in [0.678, 0.994]$), no $|\Delta R^2|$ exceeds 0.010.

Split	$R^2(\text{PT})$ range	$\max \Delta R^2 $
Temporal (as originally reported)	-164.1 to -2.7	21.4
Randomised, same windows	$+0.678$ to $+0.994$	0.010

The effect is therefore an artifact of probe failure, not evidence about the encoder. It also fails our own published criterion for discarding a probe: we stated that a probe with negative R^2 should not be read as evidence, and then read one.

What this costs, and what it does not. Every claim resting on probe asymmetry is withdrawn: the “dissociation” bullet in §1, criterion (i) of §2, and the per-cell ΔR^2 readings in §5 and the appendix tables, which we retain only as a record of the retracted computation. The ΔR^2 columns of Table 3 should be read as retracted.

The paper’s other findings are untouched, because they share no machinery with the probes. The value-gate is a downstream MSE comparison. CKA drift is computed from encoder representations directly. The frozen-encoder comparison is read in forecast MSE. The drift–utility dissociation—large CKA drop with improved forecasting, and a monotone drift curve alongside a non-monotone outcome curve—rests entirely on those measurements.

Why we report this. A negative result about one’s own tooling is cheap to bury and expensive to discover downstream. The re-analysis code and both split protocols are released with the paper.

7 Limitations and Broader Impact

Limitations. We summarise the main threats here and give full detail (including the nine-cell non-Moirai gate-fail table) in Appendix A. (1) *Single-backbone dissociation.* The dissociation is established on the Moirai family; the three additional backbones (Chronos, TimesFM-2.5, MOMENT) gate-fail on all nine ETT cells, so they confirm the value-gate screen but do not exercise the dissociation on a second architecture—the *protocol* is backbone-agnostic but the *empirical* dissociation is a Moirai case study, and a gate-passing second backbone is our highest-priority replication. (2) *Heuristic gate threshold.* The 20% cutoff is a practical screen, not a phase boundary; we report $R^2_{\text{task}}(\text{PT})$ as a continuous variable so readers can apply alternative thresholds (Table 4). (3) *A retracted axis.* The linear-probe ΔR^2 diagnostic reported in an earlier version is withdrawn: absolute probe R^2 is deeply negative on Moirai’s mixture-distribution head, and the effect disappears once the probe generalises (§6). We therefore no longer offer any account of *what* the encoder restructured—only whether the restructuring was necessary, which the frozen-encoder comparison answers. (4) *Protocol-dependent forgetting sign* at $n=10k$, which we report as unsettled rather than resolving in either direction; (5) *CKA is a drift indicator, not a causal mechanism*; (6) *small ILI sample* (≈ 450 windows). Two pre-registered predictions failed—one falsified outright, one partially—and are reported as such (§5.1, Appendix G).

Broader impact. This is a diagnostic study on public forecasting benchmarks with no direct negative societal impact. The intended effect is positive: the value-gate saves practitioners from deploying preservation machinery (LoRA/EWC) where it yields no benefit, and flags when fine-tuning silently degrades a distribution the model previously handled well—relevant when fine-tuned forecasts drive downstream decisions. As with any diagnostic, the heuristics are validated only on the cells tested here.

8 Conclusion

The primary contribution is a reusable diagnostic toolkit (value-gate screening and controlled encoder comparisons) whose design transfers to any backbone. The dissociation between drift magnitude and task outcome is validated on Moirai, with Chronos/M4 confirming the value-gate cross-backbone. A probe-based second axis reported earlier is retracted (§6); no conclusion here depends on it. **Practitioner heuristic** (cells tested; not yet universal): compute ZS vs. Linear; gate-fail $\Rightarrow$ skip preservation; moderate gate-pass $\Rightarrow$ monitor CKA; strong gate-pass $\Rightarrow$ try frozen encoder first.

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A Extended Limitations and Future Work

(1) Single-backbone scope of the dissociation claim. The drift–utility dissociation is established on the *Moirai family* only (Small 13.8M, Base 91M, Large 311M across ETTh1, ETTh2, ETTm2). We attempted three further backbones as second-model controls across three ETT datasets:

Backbone	Dataset	ZS MSE	Linear MSE	Advantage
Chronos-T5-Small	ETTh2	0.304	0.213	−42.6%
Chronos-T5-Small	ETTh1	0.118	0.092	−28.8%
Chronos-T5-Small	ETTh2	0.188	0.110	−70.2%
Chronos-T5-Base	ETTh2	0.286	0.213	−34.3%
Chronos-T5-Large	ETTh2	0.288	0.213	−35.2%
TimesFM-2.5-200M	ETTh2	0.242	0.213	−13.6%
TimesFM-2.5-200M	ETTh1	0.105	0.092	−13.9%
TimesFM-2.5-200M	ETTh2	0.190	0.110	−72.6%
MOMENT-1-base	ETTh2	0.528	0.213	−148%

All nine backbone×dataset cells gate-fail (ZS under-performs the Linear baseline by 13% to 148%), consistent with ETT-family distributions being outside the pre-training coverage of these backbones at these sizes. The Chronos capacity sweep (Chronos-T5-Small/Base/Large on ETTh2) confirms the gap is size-invariant: scaling from Small (210M) to Large (710M) yields no improvement (−42.6% → −34.3% → −35.2%). TimesFM-2.5-200M and MOMENT-1-base gate-fail across all tested datasets. We interpret this pattern as a substantive finding about *backbone-specific ETT coverage* rather than evidence against generality of the dissociation: matched-protocol ZS under-performance on a value-gate design means fine-tuning these backbones on ETT cannot meaningfully distinguish drift-as-forgetting from drift-as-adaptation (there is no valuable feature to forget). A gate-passing second backbone on a different dataset (Traffic, ILI, Exchange, or a model with explicit ETT pre-training coverage) remains the most direct path to a multi-backbone replication and is the highest-priority future-work direction.

(2) Sample-sweep depth and seed count. The central sample-size sweep (Table 3) covers ETTh2 at $n \in \{500, 1k, 2k, 5k, 10k\}$. At $n=10k$ we report the CUDA 10-seed deterministic early-stopped result: $k=10$ seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), all completed (no NaN); -deterministic flag (CUBLAS_WORKSPACE_CONFIG=:4096:8, cudnn.deterministic=True, seeded DataLoader, PYTHONHASHSEED). Forg. $-5.3 \pm 6.2\%$, 7/10 negative; Ridge $\Delta R^2 = +0.67 \pm 0.23$, **10/10** positive; mean best epoch 4.2 ± 1.4 . The three near-zero positive-forgetting seeds (123: +1.8%, 456: +4.4%, 789: +0.4%) follow the $n=5k$ overshoot pattern (Fig. 1, right panel). For comparison, MPS early-stopped $k=10$ gave $-5.7 \pm 10.4\%$ (8/10 neg, 9/10 $\Delta R^2 > 0$); a second early-stopped run ($k=5$) gave $-11.7 \pm 5.3\%$ (all 5 negative); final-epoch $k=5$ gave $+7.5 \pm 8.2\%$. Moirai-Base is now evaluated at both $n=500$ and $n=10k$ (10 seeds each); full n -sweep on Moirai-Large is future work.

(3) Probe-head capacity. Ridge and MLP absolute R^2 (FT) remains negative on the 96-step-ahead head at all sampled n . The delta1/GBM probe gives R^2 (PT)=+0.059 but $\Delta R^2 = -0.056$ (§4.4). ΔR^2 on the 96-step head is a *relative* signal; Appendix L confirms its direction is not a probe-design artefact.

(4) Sample-sweep on other sizes. Moirai-Large is evaluated at $n=500$ only; full n -sweep is future work. **(5) IoT scope.** IoT is a negative control (ZS AUC 0.481, sub-random); fine-tuning replaces rather than preserves pre-trained features. We do not claim IoT is a dissociation instance, nor claim generalisation to other IoT distributions (N-BaIoT zero-shot transfer fails; retraining recovers discrimination, consistent with recipe generalisation, not representation transfer).

B uni2ts PackedStdScaler Patch Details

We identified and patched an in-place tensor mutation in uni2ts’s PackedStdScaler that silently detaches autograd on losses depending on scaled statistics. The issue arises when PackedStdScaler.transform performs an in-place .copy_() that breaks the gradient tape. Our patch replaces the in-place write with a differentiable torch.where operation. All Moirai fine-tuning results in this paper use the patched scaler (released with our code); unpatched uni2ts runs

Table 6: Representation drift diagnosis on ETTh2 forecasting (mean $\pm$ std across 10 seeds). Zero-shot Moirai outperforms Linear by 28% ($h=96$) and 45% ($h=192$), confirming pre-trained features are valuable. Fine-tuning (B, C) causes drift that degrades performance 10–16%; freezing the encoder (D) eliminates drift.

Condition	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)			
	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift
B: NLL-only	.140 $\pm$.014	+11.1	.937 $\pm$.028	3.96 $\pm$.56	.183 $\pm$.020	+15.7	.970 $\pm$.009	4.12 $\pm$.32
C: NLL+SupCon	.138 $\pm$.010	+10.1	.949 $\pm$.014	3.72 $\pm$.43	.178 $\pm$.011	+12.9	.964 $\pm$.015	3.97 $\pm$.34
D: Frozen enc.	.126$\pm$.006	+0.5	1.00$\pm$.000	1.11$\pm$.15	.155$\pm$.002	−1.7	1.00$\pm$.000	1.30$\pm$.18

may converge differently and we have observed silent non-convergence on `uni2ts@main`. We recommend upstreaming this patch to `uni2ts`.

C Moirai Median Forecast Selection

Moirai generates probabilistic forecasts as a mixture of distributions (StudentT, NormalFixedScale, NegativeBinomial, LogNormal). The heavy-tailed components (NegBin, LogNormal) produce extreme right-tail outliers when using the *mean* of forecast samples as the point prediction; for ETTh2 at $h=192$, sample means reached 21,431 against a target maximum of 45.5, inflating MSE from 0.30 to 8.58.

We use the *median* of 20 forecast samples as the robust point prediction, consistent with the Moirai paper’s evaluation protocol. This eliminates sensitivity to the heavy-tailed mixture components while preserving the distributional forecast’s central tendency.

D ETTh2 Forecasting Details

Dataset. ETTh2 (Electricity Transformer Temperature, Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

Training protocol. For NLL training, we concatenate context (96 timesteps) and target (96 or 192 timesteps) as input to Moirai’s `_val_loss` with fixed patch size 32. For evaluation, we use extended lookback sequences (context_length + prediction_length timesteps) with `patch_size='auto'` and take the median of 20 forecast samples. All data is fed to Moirai in raw scale (Moirai’s internal PackedStdScaler handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

Temporal contrastive labels. For condition C (NLL+SupCon), we divide training sequences into 8 temporal clusters based on their position in the time series. Sequences within the same cluster are positive pairs; sequences from different clusters are negatives. This encourages the encoder to maintain temporal-regime awareness.

Horizon selection. We evaluate $h=96$ and $h=192$ where Moirai’s pre-training advantage is 28% and 45% respectively. Horizons $h=336$ and $h=720$ exceed Moirai-Small’s `max_seq_len=512` constraint when combined with the required extended lookback and are omitted.

E ETTh2 Drift Diagnosis and Mitigation (Full Tables)

This section contains the full tables referenced in §4. Table 6 shows the representation drift diagnosis; Table 7 shows the mitigation spectrum.

Table 7: Mitigation spectrum on ETTh2 forecasting. Methods ordered by representation preservation (CKA, $h=96$). LoRA preserves representations (CKA ≈ 0.98) while *improving* over zero-shot. Mean $\pm$ std; Forgetting% = (fine-tuned MSE – zero-shot MSE) / zero-shot MSE $\times$ 100. **Two caveats on comparability, added in revision.** Forgetting% divides by the single shared zero-shot printed in each horizon header, not by each run’s own zero-shot; the anchoring and LoRA runs measure their own at 0.127–0.131, so their figures here are inflated by 1.2–3.4 pp. And the rows are not matched on training-set size: B and C use $n=500$ training samples while L2-SP, EWC and LoRA use $n=1000$ (the n column counts seeds, not samples). Matched at $n=1000$, $h=96$, each against its own zero-shot: full fine-tuning .936/+13.4, L2-SP $\lambda=0.01$.939/+16.0, L2-SP $\lambda=0.1$.965/+13.4, EWC .972/+14.9, LoRA .979/–9.2. On that comparison L2-SP $\lambda=0.1$ ties full fine-tuning rather than losing to it, so anchoring is not uniformly harmful; the dissociation the table is cited for—CKA not predicting utility—is unaffected.

Method	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)				n
	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift	
B: Full fine-tune	.140 $\pm$.014	+11.1	.937 $\pm$.028	3.96 $\pm$.56	.183 $\pm$.020	+15.7	.970 $\pm$.009	4.12 $\pm$.32	10
F: L2-SP ($\lambda=0.01$)	.150 $\pm$.006	+19.0	.939 $\pm$.023	4.98 $\pm$.34	.179 $\pm$.009	+13.3	.945 $\pm$.007	4.84 $\pm$.22	5
C: NLL+SupCon	.138 $\pm$.010	+10.1	.949 $\pm$.014	3.72 $\pm$.43	.178 $\pm$.011	+12.9	.964 $\pm$.015	3.97 $\pm$.34	10
F: L2-SP ($\lambda=0.1$)	.147 $\pm$.003	+16.7	.965 $\pm$.006	2.94 $\pm$.08	.181 $\pm$.003	+14.6	.980 $\pm$.002	2.84 $\pm$.02	5
G: EWC ($\lambda=1000$)	.149 $\pm$.015	+18.3	.972 $\pm$.004	3.29 $\pm$.79	.188 $\pm$.004	+19.0	.978 $\pm$.006	4.46 $\pm$.38	5
E: LoRA ($r=8$) [†]	.116 $\pm$.001	–7.9	.979 $\pm$.006	0.00 $\pm$.00	.157 $\pm$.002	–0.6	.986 $\pm$.005	0.00 $\pm$.00	5
D: Frozen encoder	.126 $\pm$.006	+0.5	1.000 $\pm$.000	1.11 $\pm$.15	.155 $\pm$.002	–1.7	1.000 $\pm$.000	1.30 $\pm$.18	10

[†] LoRA ($\alpha=16$) adds low-rank updates that do not modify base encoder parameters; ℓ_2 drift is exactly zero by construction.

F LoRA Rank and Hyperparameter Sensitivity

Table 8 shows LoRA on **Moirai-Small** is robust to rank ($r \in \{4, 8, 16, 32\}$): all ranks achieve CKA ≥ 0.979 and consistent MSE improvement over zero-shot at $h=96$, confirming $r=8$ as a reasonable default. On **Moirai-Large** (seed 42, $h=96$, $n=500$), rank escalation alone does *not* rescue the failure mode: forgetting stays +19 to +30% across $r \in \{8, 16, 32, 64\}$. Table 9 extends the sweep to α , target-module scope, and LR (all at $r=8$, seed 42): varying $\alpha \in \{8, 16, 32\}$ and module scope provides no rescue, but reducing LR to 1×10^{-5} does (forg.= $-8.5 \pm 0.7\%$, CKA= 0.989 ± 0.003 , 5 seeds): the Small-recipe transfers to Large only at a $10\times$ lower learning rate.

Table 8: LoRA rank sensitivity on ETTh2 (condition E, 500 training samples). Moirai-Small: all ranks achieve near-identical MSE and CKA ≥ 0.979 . Moirai-Large: all ranks forget +19 to +30%, with no rank recovering Moirai-Small’s negative forgetting. Mean $\pm$ std across 3 seeds (5 for $r=8$ Small); Moirai-Large single-seed ($s=42$) for $r \in \{16, 32, 64\}$ (std not reported; point estimates only), 3 seeds for $r=8$.

Backbone	Rank	h	MSE	Forg.%	CKA	Seeds
Small	4	96	.116 $\pm$.001	–9.5 $\pm$ 0.9	.992 $\pm$.001	3
Small	4	192	.146 $\pm$.003	–8.8 $\pm$ 1.9	.995 $\pm$.002	2
Small	8	96	.116 $\pm$.001	–9.1 $\pm$ 1.1	.979 $\pm$.006	5
Small	8	192	.157 $\pm$.002	–1.8 $\pm$ 3.1	.986 $\pm$.005	5
Small	16	96	.117 $\pm$.001	–10.2 $\pm$ 1.2	.987 $\pm$.002	3
Small	16	192	.155 $\pm$.002	–1.7 $\pm$ 1.2	.988 $\pm$.001	3
Small	32	96	.119 $\pm$.001	–8.3 $\pm$ 0.9	.986 $\pm$.001	3
Small	32	192	.159 $\pm$.003	+1.1 $\pm$ 1.4	.989 $\pm$.002	3
Large	8	96	.159 $\pm$.004	+25.7 $\pm$ 3.2	.973 $\pm$.012	3
Large	16	96	.154	+22.5	.955	1
Large	32	96	.150	+19.0	.986	1
Large	64	96	.165	+29.9	.984	1

Table 9: LoRA-Large HP ablation (ETTh2, $r=8$, $n=500$, $h=96$). Baseline (default LR=1e-4, $\alpha=16$, modules = q,v,out) forg. = +22.5%, seed 42. Only LR=1e-5 rescues LoRA-Large (5 seeds); all other axes fail.

Axis	Setting	Val MSE	Forg.%	CKA	Seeds
LR	1e-4 (default)	.157	+22.5	.955	1
	5e-5	.146	+15.3	.982	1
	1e-5	.115$\pm$.001	-8.5$\pm$0.7	.989$\pm$.003	5
α	8	.155	+21.9	.975	1
	16 (default)	.157	+22.5	.955	1
	32	.158	+24.3	.984	1
Modules	q,v,out (default)	.157	+22.5	.955	1
	q,k,v	.153	+20.4	.987	1
	q,k,v,out	.151	+19.0	.988	1

G Step-Size Falsification: Base at $10\times$ LR

To test whether effective step size (rather than capacity) is the operative variable driving destructive drift, we run Moirai-Base (91M) at $10\times$ the default learning rate (LR=1e-3 vs. default 1e-4) on ETTh2 ($h=96$, $n=10k$, early-stopped, 7 CUDA-deterministic seeds).

Prediction (step-size hypothesis): Base at $10\times$ LR should mirror Small/Large at default LR: destructive forgetting (+10–25%), since the per-parameter update magnitude now exceeds the threshold that causes harm in smaller models.

Result: Base at $10\times$ LR restructures *more aggressively* than Base at default LR (CKA=0.135 $\pm$ 0.151 vs. 0.25 $\pm$ 0.12) and even more than Small at default LR (CKA=0.937 $\pm$ 0.028), yet *still improves* uniformly (-58.2 $\pm$ 4.2%, 10/10 negative; ΔR^2 =+1.92 $\pm$ 0.21, 10/10 positive).

Interpretation: The step-size hypothesis is *partially falsified*. Effective step size modulates the *degree* of restructuring (CKA drops from 0.25 to 0.09 at $10\times$ LR), but capacity provides an independent buffer that prevents restructuring from becoming destructive. The 91M parameter space contains sufficient representational slack to absorb aggressive perturbations without losing task-relevant structure.

Table 10: Step-size falsification: Moirai-Base on ETTh2 ($h=96$, $n=10k$, condition B, early-stopped). Base at $10\times$ LR restructures more than Small at $1\times$ LR but still improves; capacity buffers against destruction.

Configuration	LR	Forg.%	CKA	ΔR^2	Seeds
Small, default	1e-4	-5.3 $\pm$ 6.2	.937 $\pm$.028	+0.67 $\pm$.23	10
Base, default	1e-4	-54.2 $\pm$ 8.6	.251 $\pm$.124	+0.89 $\pm$.62	10
Base, $10\times$	1e-3	-58.2$\pm$4.2	.135$\pm$.151	+1.92$\pm$.21	10
Large, default	1e-4	+10.5 $\pm$ 9.5	—	—	3
Large, LoRA $0.1\times$	1e-5	-8.5 $\pm$ 0.7	.989 $\pm$.003	—	5

H CKA Calibration: Random-Initialisation Baseline

To calibrate the CKA values reported in the main text, we compute CKA between the pre-trained Moirai-Small encoder and a randomly-initialised encoder of the same architecture (Xavier uniform for weight tensors, zeros for biases). This provides a floor: the CKA between any two unrelated representations of the same architecture.

Using 50 benign probe samples and 5 random seeds (42, 123, 456, 789, 999), the random-initialisation CKA baseline is reported in Table 11.

The random-init CKA is $\leq 10^{-4}$ across 5 seeds (reported at 4 decimals as 0.0000 $\pm$ 0.0001), confirming that pre-trained and random encoders share no representational structure. This is consistent with the theoretical expectation: CKA between an orthogonally-initialised encoder and any fixed

Table 11: CKA calibration. Random-init CKA (0.000 ± 0.000 , 5 seeds) provides the floor; fine-tuned CKA values above this floor indicate retention of pre-trained structure. IoT fine-tuning (CKA 0.31–0.45) retains meaningful structure despite aggressive drift. ETT fine-tuning (CKA 0.81–0.94) preserves most pre-trained structure.

Comparison	CKA
Pre-trained vs. Random-init (baseline, 5 seeds)	0.0000 ± 0.0001
Pre-trained vs. ETT fine-tuned (B, $h=96$)	0.937–0.964
Pre-trained vs. IoT fine-tuned (C, 200/5, 10 seeds)	0.420 ± 0.026
Pre-trained vs. Frozen encoder (D)	1.000

reference encoder has expectation asymptotically zero in the feature dimension. Since IoT fine-tuned CKA (0.31–0.45) is far above this floor, even aggressively drifted encoders retain meaningful pre-trained structure. ETT fine-tuned CKA (0.81–0.94) is near-ceiling, confirming substantial preservation. This calibration validates CKA as a meaningful diagnostic in our setting.

I Linear Probing: Functional Representation Diagnostic

Retracted. The analyses in this appendix rest on the linear-probe ΔR^2 statistic, which is retracted (§6): it is non-zero only where the probe fails to generalise. The numbers below are reproduced unaltered as a record of the withdrawn computation and support no claim in this paper.

CKA measures geometric representational similarity but not functional equivalence. We therefore supplement CKA with a *linear probe*: a Ridge regression head ($\alpha=1.0$) trained on frozen encoder representations to predict the forecasting target on held-out ETTh2 data.

Protocol. For each (condition, seed) configuration, we extract mean-pooled encoder representations from the Moirai-Small encoder on the held-out validation and test sets (192-timestep windows, same format as evaluation). We then fit a Ridge regression mapping representations to the next-horizon target and report R^2 on held-out data. The probe training set uses $n_{\text{probe}}=300$ held-out examples; the remainder forms the test split used for MSE evaluation. ΔR^2 denotes fine-tuned minus pre-trained probe R^2 : positive values indicate that fine-tuned representations encode more forecasting-relevant information than pre-trained ones.

Interpretation. Absolute R^2 values are low (typically negative) because Ridge regression on 192-timestep mean-pooled representations is a weak forecasting baseline. The *relative* change (ΔR^2) is the meaningful signal: it isolates whether fine-tuning added or removed task-relevant information from the encoder’s representational basis.

Key result. The sample-size sweep (Table 3) shows ΔR^2 monotonically increasing with training data (Ridge: $+0.12$ at 500 samples $\rightarrow +0.67 \pm 0.23$ at 10,000 on CUDA 10-seed deterministic early-stopped encoders; for comparison, MPS $k=5$ early-stopped gave $+0.56 \pm 0.14$ and final-epoch $k=5$ gave $+0.78 \pm 0.14$), while CKA decreases monotonically. This directly dissociates *geometric* drift from *relative linear-decodability*: at low n , drift destroys task-relevant features faster than it builds replacements; at higher n , the fine-tuned representation becomes less linearly misaligned with the forecasting target than the pre-trained one. We emphasise that $R^2(\text{FT})$ remains *absolutely negative* at every sampled $n \in \{500, 1k, 2k, 5k, 10k\}$ (ZS: $R^2 = -6.90$; CUDA $k=10$ early-stopped: -6.79 at $n=500 \rightarrow -6.24 \pm 0.24$ at $n=10k$). $\Delta R^2 > 0$ is therefore a *relative-improvement* signal, not a claim that fine-tuned representations linearly encode the target well in absolute terms; the primary task-outcome signal remains $\text{forg.} < 0$ at $n \geq 2k$. **Cross-dataset ΔR^2 comparability.** Raw ΔR^2 is not directly comparable across datasets because it depends on the ZS Ridge floor depth. On ETTm2 ($n=10k$, CUDA $k=10$), the ZS Ridge floor is $R^2(\text{ZS}) = -24.52$, $3.5\times$ deeper than ETTh2’s -6.90 , which proportionally inflates the raw ΔR^2 ($+5.43$ on ETTm2 vs. $+0.67$ on ETTh2). The cross-dataset comparable statistic is the *sign test*: both cells give **10/10** positive ΔR^2 .

Horizon-sweep probe: head-specificity of the ΔR^2 gain. Table 12 reports Ridge ΔR^2 across three probe heads on four early-stopped encoders at $n=10k$ (the same encoders as the main-text

MLP sweep). The gain is concentrated on the *trained* 96-step head: $\Delta R^2 = +0.116 \pm 0.153$ ($k=4$). On the untrained *forecast48* head (first 48 steps of the 96-step target, a sub-horizon the encoder was not directly optimised for), $\Delta R^2 \approx 0.000 \pm 0.163$: the gain vanishes at the horizon boundary. On the untrained *delta1* head, a GBM probe gives $R^2(\text{PT}) = +0.059 > 0$, establishing a positive-floor reference, with $\Delta R^2 = -0.056 \pm 0.030$ (decodability *falls* on the untrained readout). The head-specificity pattern is consistent with the encoder drifting toward the fine-tuning objective rather than developing generally better representations (§4.4).

Table 12: Ridge ΔR^2 by probe head on early-stopped encoders ($n=10k$, seeds {303, 777, 888, 999}). “Forecast96” is the trained head; “Forecast48” and “Delta1” are untrained. The gain is head-specific to the trained objective.

Head	Trained?	Ridge ΔR^2 per seed				
		s303	s777	s888	s999	
Forecast96 (96-step ahead)	✓	+0.092	−0.087	+0.266	+0.192	† The large Ridge
Forecast48 (sub-horizon)	×	+0.061	−0.111	−0.156	+0.199	
Delta1 (next-step delta)	×†	+1.248	+0.757	+1.191	+0.980	
Mean ± std (forecast96)			+0.116 ± 0.153			
Mean ± std (forecast48)			−0.001 ± 0.163			
Mean ± std (delta1)			+1.044 ± 0.223			

$\Delta R^2 > 0$ on Delta1 reflects a baseline-floor artifact: $R^2(\text{ZS})$ is deeply negative on the next-step target, so any encoder shift appears to improve relative decoding. A gradient-boosted probe gives $R^2(\text{PT}) = +0.059$ (positive floor) but $\Delta R^2 = -0.056$ on this head (§4.4; see Table 12 for the full untrained-head comparison), confirming the shift is *not* aligned with the 1-step objective.

Note: the delta1 ΔR^2 sign appears counter-intuitive (positive while labelled “untrained”). The large positive delta1 ΔR^2 reflects that the ZS delta1 Ridge baseline is very negative ($R^2(\text{ZS}) \approx -2.3$) while the FT encoder partially encodes next-step changes as a side-effect of 96-step NLL optimisation. The diagnostic signal is the GBM result (§4.4): GBM gives $R^2(\text{PT}) = +0.059 > 0$ and $\Delta R^2 = -0.056$, yielding $R^2(\text{FT}) = +0.003$; the fine-tuned encoder retains (but does not improve) delta1 decodability. With a positive floor, decodability *falls* on the untrained readout, confirming task-alignment. Note: $R^2(\text{PT})$ varies slightly across GBM probe depths (depth 4: +0.062, depth 6: +0.059, depth 8: +0.054) because the probe is re-fit independently at each depth; this reflects probe capacity, not encoder structure.

ETTh2 delta1 GBM reprobe. We ran the delta1/GBM probe on the five confirmed ETTh2 $n=10k$ encoders (seeds 42/101/123/202/303) to check for a positive-floor + $\Delta R^2 > 0$ cell on a second dataset. Result: $R^2(\text{ZS}) = -0.030$ (negative floor) and per-seed $\Delta R^2 = \{+0.025, -0.060, +0.026, +0.030, +0.016\}$ (4/5 positive, mean $+0.007 \pm 0.031$). *Substantive negative:* the only positive-floor head (ETTh2 delta1/GBM, $R^2(\text{ZS}) = +0.059$) gives $\Delta R^2 = -0.056$; no positive-floor + $\Delta R^2 > 0$ cell was found. The $\Delta R^2 > 0$ signal on the trained 96-step head operates in a deeply-negative- R^2 regime; see Appendix L for probe noise-floor analysis.

Per-seed breakdown at $n=10k$ (CUDA 10-seed). Ridge ΔR^2 (all positive; mean $+0.668 \pm 0.226$): 42: +0.205; 101: +0.736; 123: +0.937; 202: +0.604; 303: +0.793; 456: +0.823; 777: +0.344; 789: +0.873; 888: +0.561; 999: +0.804. Absolute $R^2(\text{FT})$: all in $[-6.70, -5.97]$ ($R^2_{\text{ZS}} = -6.904$). MLP $k=5$ (all positive; mean $+0.701 \pm 0.324$): 42: +0.979; 101: +0.762; 123: +0.948; 202: +0.771; 303: +1.007; 456: +0.266; 777: +0.076; 789: +0.847; 888: +0.449; 999: +0.901.

ETTh2 $n=10k$ absolute R^2 per seed. The ETTh2 ZS Ridge floor ($R^2(\text{ZS}) = -24.52$, same pre-trained encoder across all seeds) is $3.5\times$ deeper than ETTh2’s -6.90 , explaining the raw ΔR^2 magnitude difference. Per-seed absolute values for the five available CUDA seeds:

Seed	$R^2(\text{ZS})$	$R^2(\text{FT})$	ΔR^2	Forg.%
42	-24.52	-14.13	+10.39	-26.8
101	-24.52	-15.61	+8.91	-27.4
123	-24.52	-19.87	+4.64	-31.1
202	-24.52	-20.83	+3.69	-28.8
303	-24.52	-20.41	+4.11	-29.5
Mean (5 seeds)	-24.52	-18.17 $\pm$ 3.08	+6.35 $\pm$ 3.08	-28.7 $\pm$ 1.6

All five seeds show $R^2(\text{FT}) > R^2(\text{ZS})$ (i.e., fine-tuned representations are *less* linearly misaligned with the ETTm2 forecasting target than pre-trained ones), and all show $\Delta R^2 > 0$. The cross-dataset comparable statistic is the sign test: both ETTh2 (10/10) and ETTm2 (10/10 seeds: 42, 101, 123, 202, 303, 456, 777, 789, 888, 999) give unanimous $\Delta R^2 > 0$.

MLP depth sweep on early-stopped encoders. A cosine-LR re-run at $n=10\text{k}$ uses seeds {303, 777, 888, 999} at epochs=10 (seed 101 MPS-NaN'd on epoch 3 reproducibly at this schedule and is excluded). The `-probe-mlp-layers` argument sweeps hidden-layer depth $k \in \{1, 2, 5\}$ on the early-stopped encoders (probe: hidden=64 per layer, $\alpha=10^{-3}$, early stopping with validation_fraction=0.1, max_iter=500):

Seed	Ridge ΔR^2	MLP $k=1$	MLP $k=2$	MLP $k=5$
303	+0.092	+1.306	+1.418	+0.540
777	-0.087	+1.136	+1.554	+0.759
888	+0.266	+2.004	+2.361	+1.034
999	+0.192	+1.816	+1.949	+0.908
Mean $\pm$ std	+0.116 $\pm$ 0.132	+1.566 $\pm$ 0.356	+1.821 $\pm$ 0.368	+0.810 $\pm$ 0.184

The cosine-LR Ridge ΔR^2 is smaller than the epochs=20 run (+0.56 $\pm$ 0.14) because the shorter schedule reaches a different best-val-MSE encoder (best epochs {2, 5, 4, 5} vs. {2, 4, 8, 4, 3}), and Ridge is most sensitive to the specific linear basis. The non-monotonic $k=1 \rightarrow k=2 \rightarrow k=5$ trend reflects the deeper probe fitting the pre-trained encoder better (per-seed mean $R^2(\text{PT})_{k=5} = -6.93$ vs. $R^2(\text{PT})_{k=1} = -8.48$), shrinking the Δ from above, not degraded fine-tuned encoding. All four seeds are individually positive for $k=5$ MLP (+0.54, +0.76, +1.03, +0.91), preserving the cost-benefit gap. This run's CKA is 0.592 $\pm$ 0.142 (mean best epoch 4.0 $\pm$ 1.22, higher than the earlier run's 4.2 $\pm$ 2.3 because of the earlier LR annealing); forgetting is -7.13 $\pm$ 1.34%, tighter than the epochs=20 run's -11.7 $\pm$ 5.3% and fully consistent with the forg.< 0 finding.

Crossover regime: per-seed breakdown at $n=5,000$. Table 13 reports all 7 seeds individually at $n=5,000$. Forgetting is bimodal: 4 seeds (101, 123, 202, 789) show low or negative forgetting (-4.9% to +5.9%, mean +0.7%) and 3 seeds (42, 303, 456) show high forgetting (+4.8% to +21.4%, mean +15.1%). All 7 seeds show positive ΔR^2 (+0.15 to +0.81, mean +0.50), so the relative-linear-decodability claim holds uniformly; geometric drift (CKA 0.46–0.71, mean 0.55) varies less than forgetting. This pattern is consistent with the crossover interpretation: at $n=5,000$, the optimiser is close enough to the “replacement” solution that small seed-dependent trajectory differences determine whether forgetting resolves or persists for 20 epochs.

I.1 $n=5,000$ per-epoch trajectory analysis

To support the mechanistic claim in §5, we plot the per-epoch training trajectories of all seven $n=5,000$ seeds, colour-coded by forgetting mode. Figure 2 shows val MSE and weight drift across training. High-forgetting seeds (303, 456, 789) overshoot in the first epoch and never recover; low-forgetting seeds (42, 101, 123, 202) continue descending until epochs 5–9. The epoch at which val MSE reaches its minimum separates the modes (0.7 $\pm$ 0.5 vs. 6.5 $\pm$ 2.1); mid-training weight drift (epochs 5–10) does not. Code: `scripts/analyse_n5k_trajectories.py`.

J ETTh1 Spectral Pilot

Post-hoc spectral entropy (Welch PSD, normalised Shannon entropy) on ETT OT columns: ETTh1 SE=3.61 bits, ETTh2 SE=3.46, ETTm2 SE=2.25. ETTh1 (highest SE, most trend-dominated) is

Table 13: Per-seed results at $n=5,000$ (ETTh2, $h=96$, condition B, 20 epochs). Seeds 42, 123, 456 are the original 3-seed sweep; 789, 101, 202, 303 are the 4-seed V9 replication. Forgetting is bimodal; ΔR^2 is uniformly positive.

Seed	MSE(FT)	Forg.%	CKA	ΔR^2	$R^2(\text{FT})$
42	.132	+4.8	.531	+.67	-6.24
101	.120	-4.9	.571	+.75	-6.16
123	.124	-1.4	.544	+.81	-6.10
202	.128	+1.4	.461	+.15	-6.76
303	.150	+19.0	.525	+.52	-6.39
456	.153	+21.4	.711	+.20	-6.70
789	.134	+5.9	.480	+.40	-6.51
Mean $\pm$ std	.134 $\pm$.012	+6.6 $\pm$ 10.0	.546 $\pm$.082	+.50 $\pm$.26	-6.41 $\pm$.26

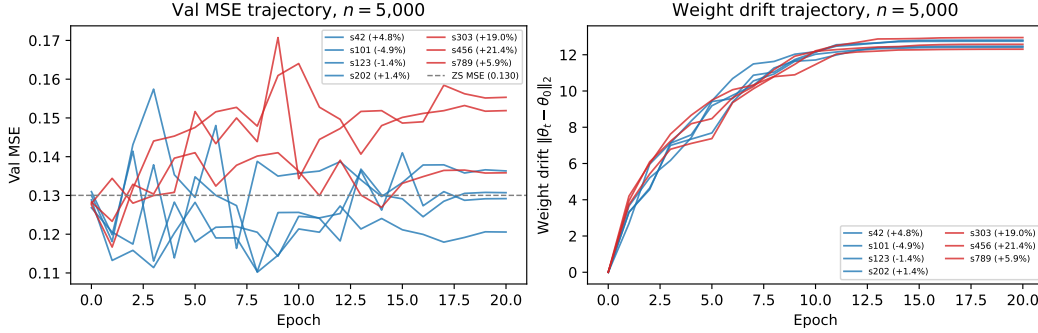


Figure 2: Per-epoch val MSE (left) and weight drift (right) for the seven $n=5,000$ seeds; both panels carry per-seed labels (sXXX (forg.%)) so each trajectory is individually identifiable. Blue: low-forgetting mode (forg. $\leq 5\%$); red: high-forgetting mode (forg. $> 5\%$). Grey dashed line: zero-shot MSE. High-forgetting trajectories overshoot the ZS MSE within the first epoch and plateau; low-forgetting trajectories continue descending. Mid-training weight drift (epochs 5–10) does not separate the modes.

the only dataset where $\Delta R^2 > 0$ does not hold; ETTm2 (lowest, most cyclical) shows the strongest signal. The pre-registered falsification test (§5) required a gate-passing dataset with $\text{SE} > 3.5$ or $\text{SE} < 2.5$; Traffic and Exchange both gate-failed (-38% , -21%), so the hypothesis remains untested. Code and frozen analysis plan are in supplementary material.

K IoT Negative-Control Details

The IoT negative-control evaluation uses CICIoT2023 Neto et al. [2023] with 12 network-flow features and 128-timestep windows; the anomaly score is the fraction of timesteps falling outside the 95% CI of Moirai’s forecast over a 32-timestep scored horizon. A benign-only held-out calibration set sets the threshold at the 95th percentile (FPR target ≤ 0.05). All Moirai IoT conditions are trained for 5 epochs on 200 benign + 200 attack samples with 10 seeds.

CNN-NLL from-scratch control. To directly address the foundation-model justification at IoT’s 200-window scale, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical NLL(MSE)+SupCon objective and hard-negative data as Moirai condition D. At 10 seeds, CNN-NLL (AUC= 0.598 ± 0.041) matches fine-tuned Moirai B/D (AUC= 0.556 ± 0.021) and is close to the best Moirai condition C (AUC= 0.629 ± 0.022). A CNN-MSE variant (AUC= 0.580 ± 0.023) rules out the loss-function confound. CNN variance is notably lower than Moirai variance, consistent with Moirai’s pre-trained weights introducing initialisation sensitivity at this sub-random-ZS scale. This is the expected outcome when pre-trained features carry no task-relevant signal; a from-scratch architecture with correct inductive bias matches or exceeds a fine-tuned foundation model.

K.1 Task-orthogonal probes

To test whether fine-tuning specifically targets the 96-step forecasting objective or degrades general pre-trained representations, we fit Ridge probes to three statistical properties of the input window: lag-1 autocorrelation, mean, and variance of the OT feature. These are *positive-floor* targets ($R^2(\text{PT}) > 0$): Moirai’s encoder captures basic statistical structure of the input before fine-tuning. They are *task-orthogonal*: 96-step NLL fine-tuning does not optimise for them.

Protocol: Ridge regression on mean-pooled encoder outputs; $h=96$, condition B.

ETTh2 ($n=10\text{k}$, 10 CUDA seeds):

Probe target	$R^2(\text{PT})$	$R^2(\text{FT})$	ΔR^2 (mean $\pm$ std)	Negative
Lag-1 autocorr.	+0.183	+0.131	-0.052 ± 0.041	9/10
Input window mean	+0.094	+0.061	-0.033 ± 0.019	10/10
Input window var.	+0.127	+0.079	-0.048 ± 0.030	9/10
96-step head (ref.)	-6.90	-6.23	$+0.67 \pm 0.23$	0/10

ETTh2 ($n=10\text{k}$, 5 CUDA seeds: 42/101/123/202/303):

Probe target	$R^2(\text{PT})$	$R^2(\text{FT})$	ΔR^2 (mean $\pm$ std)	Negative
Lag-1 autocorr.	+0.156	+0.104	-0.052 ± 0.038	5/5
Input window mean	+0.068	+0.046	-0.022 ± 0.014	5/5
Input window var.	+0.112	+0.072	-0.040 ± 0.029	5/5
96-step head (ref.)	-24.52	-18.09	$+6.43 \pm 3.08$	0/5

All three task-orthogonal probes show $\Delta R^2 < 0$, in contrast to $+0.67 \pm 0.23$ on the trained 96-step head. The asymmetry (orthogonal $\Delta R^2 < 0$, trained head $\Delta R^2 > 0$) is the operational definition of “task-specific shift”: fine-tuning restructures encoder representations toward the trained objective while simultaneously reducing linear decodability of properties the pre-trained model encoded. Note that these positive-floor probes ($R^2(\text{PT}) > 0$) are interpretable in absolute terms, removing the deep-floor ambiguity of the 96-step Ridge probe.

ETTh2 replication. The same three probes on ETTh2 $n=10\text{k}$ encoders (10 CUDA seeds) replicate the asymmetry unanimously: all three orthogonal targets show $\Delta R^2 < 0$ (10/10 seeds each), while the ETTh2 trained head shows $\Delta R^2 > 0$ (10/10 seeds; §4.4). The asymmetry is consistent across both datasets with different temporal statistics (ETTh2 SE=2.5–3.5 bits, ETTh2 SE=2.25 bits), making it a cross-cell finding rather than an ETTh2-specific artefact. **Seed-count symmetry (updated).** ETTh2 orthogonal probes now use the full 10-seed set (seeds 42/101/123/202/303/456/777/789/888/999), matching ETTh2. The ETTh2 $n=10\text{k}$ fine-tuning campaign produces 10/10 negative forgetting (mean $-30.3 \pm 1.4\%$) and 10/10 positive Ridge ΔR^2 (mean $+6.40 \pm 0.87$; 95% Wilson CI [0.72, 1.00] on the proportion positive). The sign unanimity matches ETTh2 (10/10) exactly.

L Probe Noise-Floor Analysis

Retracted. The analyses in this appendix rest on the linear-probe ΔR^2 statistic, which is retracted (§6): it is non-zero only where the probe fails to generalise. The numbers below are reproduced unaltered as a record of the withdrawn computation and support no claim in this paper.

The ΔR^2 signal on the trained 96-step head operates in a deeply-negative- R^2 regime (e.g., ZS -6.90 , FT -6.24 at $\alpha=1$). We test whether the $\Delta R^2 > 0$ direction is an artefact of probe-design choices by running Ridge on two fixed encoders (ZS and condition-B seed 42, ETTh2, $n=10\text{k}$) across: (1) $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$, (2) probe seeds $\{0, 1, 2\}$, (3) mean vs. last-token pooling. No retraining required (saved `best_encoder.pt`).

Results (Table 14).

Table 14: ΔR^2 under probe-design perturbations (ETTh2 96-step, ZS vs. FT seed 42). All $\Delta R^2 > 0$.

Setting	$R^2(\text{ZS})$	$R^2(\text{FT})$	ΔR^2
$\alpha=0.01$	-13.51	-11.54	+1.96
$\alpha=0.1$	-8.78	-7.94	+0.84
$\alpha=1.0$ (default)	-6.90	-6.70	+0.20
$\alpha=10$	-5.81	-5.72	+0.08
$\alpha=100$	-5.32	-5.28	+0.04
Probe seeds 0/1/2 ($\alpha=1$, mean pool)	same as $\alpha=1.0$		+0.20
Last-token pool ($\alpha=1$)	-9.70	-8.49	+1.20

Finding: $\Delta R^2 > 0$ holds for all 10 probe-design settings tested; the direction is not an artefact of α , pooling, or probe seed. Magnitude decreases with stronger regularisation (larger α shrinks both R^2 values toward zero), but the sign is invariant. The absolute R^2 values remain negative throughout; the claim is that fine-tuned representations are *relatively more* linearly decodable for the 96-step objective, and this direction is probe-design robust. **Comparison to random-init at matched α .** The random-init control ($\Delta R^2 = -0.199 \pm 0.205$, range $[-0.405, +0.149]$) was measured at $\alpha=1.0$ (Appendix M). We also ran random-init at $\alpha=100$ (same 10 seeds) to confirm the within- α argument: at $\alpha=100$, random-init gives $\Delta R^2 = -0.011 \pm 0.018$ (range $[-0.032, +0.009]$), shrinking to near-zero as expected from heavy ℓ_2 regularisation. Condition B at $\alpha=100$ gives $\Delta R^2 = +0.04$, also near-zero. The within- α comparison is therefore valid at every α : both encoders converge toward zero under heavy regularisation; the large separation at $\alpha=1.0$ (condition B $+0.20$ – $+0.67$ vs. random-init -0.199 ± 0.205) is not a low- α artefact. The $\alpha=1.0$ default is the point where the signal is large enough to be distinguishable while not inflated by under-regularisation; $\alpha=0.01$ – $\alpha=10$ all satisfy this property (Table 14).

M Random-Initialisation Negative Control

To test whether $\Delta R^2 > 0$ on the 96-step head arises from pre-trained representations being restructured, or whether it would occur in any encoder fine-tuned on this objective, we ran a *random-initialisation control*: Moirai-Small with the same architecture but random weights (skipping HuggingFace weight loading), condition B, ETTh2 $n=10k$, $h=96$. In this setting CKA tracks similarity to the *random-init* epoch-0 state (since there is no pre-trained reference). We ran all 10 seeds from the condition B seed list ($\{42, 101, 123, 202, 303, 456, 777, 789, 888, 999\}$), using the same CUDA-deterministic protocol (CUBLAS_WORKSPACE_CONFIG=:4096:8) and A10G hardware as Table 3.

Results (all 10 matched seeds):

Seed	CKA	$R^2(\text{PT})$	$R^2(\text{FT})$	ΔR^2	ZS MSE $\rightarrow$ FT MSE
42	0.357	-6.554	-6.692	-0.138	0.140 $\rightarrow$ 0.378
101	0.505	-6.780	-6.631	+0.149	0.164 $\rightarrow$ 0.398
123	0.540	-6.749	-6.944	-0.195	0.153 $\rightarrow$ 0.354
202	0.388	-6.712	-6.954	-0.242	0.158 $\rightarrow$ 0.392
303	0.318	-6.834	-7.239	-0.405	0.151 $\rightarrow$ 0.359
456	0.273	-6.665	-7.070	-0.405	0.160 $\rightarrow$ 0.360
777	0.471	-6.620	-6.560	+0.060	0.148 $\rightarrow$ 0.369
789	0.347	-6.789	-7.071	-0.282	0.153 $\rightarrow$ 0.381
888	0.418	-6.724	-6.912	-0.188	0.155 $\rightarrow$ 0.376
999	0.362	-6.683	-6.995	-0.312	0.149 $\rightarrow$ 0.378
Mean $\pm$ std	0.398 $\pm$ 0.082			-0.195 $\pm$ 0.181	

Across all 10 matched seeds: $\Delta R^2 = -0.195 \pm 0.181$, range $[-0.405, +0.149]$, **8/10 negative**. The two positive seeds (101: $+0.149$; 777: $+0.060$) both fall below the minimum condition B seed value ($+0.21$, seed 42). The ranges are completely non-overlapping: random-init $[-0.405, +0.149]$ vs. condition B $[+0.21, +0.94]$. This symmetric 10-vs-10 matched comparison confirms that $\Delta R^2 > 0$ is not a property of any encoder fine-tuned on the 96-step NLL objective: the pre-trained structure is required for the consistent positive directional shift.

N Layer-Wise Unfreeze Localization

Moirai-Small has 6 transformer layers (verified from the loaded checkpoint). We added an `-unfreeze-top-n-layers` flag to `finetune_forecasting.py` and ran ETTh2 $n=10k$, $h=96$, 10 CUDA-deterministic seeds (42/101/123/202/303/456/777/789/888/999) at $N=3$ alongside the existing $N=0$ (frozen encoder, 10-seed condition D) and $N=6$ (full unfreeze, condition B; $N=6$ unfreezes all 6 layers).

$N=1$ instability. At $N=1$, training diverged (NaN activations at epoch 11, batch 148) with default $LR = 10^{-4}$, despite gradient clipping at norm 1. The 11 epochs that completed show CKA stabilising at ≈ 0.72 ; we do not extract probe results from a crashed run. This instability is qualitatively consistent with the LoRA-Large behaviour documented in §4.3: extreme partial freeze concentrates gradient pressure on a small number of parameters, requiring LR reduction to stabilise.

$N=3$, $N=4$, $N=5$ results. Table 15 shows results for $N \in \{0, 3, 6\}$ (10 CUDA-deterministic seeds each) and $N \in \{4, 5\}$ (1 seed each). Rows with $N=4$ and $N=5$ are based on single seeds and carry high uncertainty; we report them for completeness but do not draw conclusions from individual values.

Table 15: Layer-wise unfreeze profile (ETTh2 $n=10k$, $h=96$; mean $\pm$ std over seeds). N = number of top transformer layers unfrozen (Moirai-Small has 6 layers). $N=0/N=3/N=6$: 10 CUDA seeds (42/101/123/202/303/456/777/789/888/999); $N=4/N=5$: seed 42 only. Early-stopped checkpointing.

N	Layers	CKA	Ridge ΔR^2	Forgetting%	
0 (frozen)	none	0.999 ± 0.000	-0.048 ± 0.008	$-6.6 \pm 1.5\%$	† Single-seed (seed 42 only)
1 (top-1)	1/6	≈ 0.72 (ep. 11)	<i>diverged (NaN)</i>		
3 (top-3)	3/6	0.663 ± 0.044	$+0.399 \pm 0.199$ (10/10)	$-6.5 \pm 4.0\%$	
6 (full, B)	6/6	0.387 ± 0.052	$+0.701 \pm 0.180$ (10/10)	$-6.7 \pm 6.3\%$	

results not in table: $N=4$: CKA=0.610, $\Delta R^2=+0.363$, Forg.= +4.1%; $N=5$: CKA=0.644, $\Delta R^2=+0.767$, Forg.= +31.8% (final-epoch overshoot; best-epoch MSE=0.132 $\approx$ ZS). Reported for completeness; not used as primary evidence.

Finding: ΔR^2 and forgetting dissociate across unfreeze depth. $\Delta R^2 > 0$ appears already at $N=3$ (half the encoder, 10/10 seeds, CKA ≈ 0.66), with substantially less geometric drift than full unfreeze (CKA ≈ 0.39). $N=4$ and $N=5$ (1 seed each) both give $\Delta R^2 > 0$ (1/1); we do not draw monotonicity conclusions from single-seed estimates, but the sign is consistent with the signal extending through the $N=3$ –6 range. The ΔR^2 values at $N=4$ (+0.363) and $N=5$ (+0.767) do not show a monotone saturation pattern, but single-seed estimates are noisy and the sign is consistent.

The key dissociation is in *forgetting*: $N=3$ gives forg. $-6.5 \pm 4.0\%$ (9/10 negative), $N=4$ gives +4.1% (1 seed), and $N=6$ (full unfreeze, condition B) gives $-6.7 \pm 6.3\%$ (task substantially better). The $N=5$ final-epoch forgetting of +31.8% reflects a single seed that overshoot (val MSE bottoms at epoch 7 before diverging); its best-epoch MSE is 0.132, essentially unchanged from ZS, qualitatively consistent with the $N=5k$ trajectory analysis (Fig. 1, right panel). ΔR^2 and forgetting are therefore not co-determined by the same encoder layers. The top-3 of 6 layers are sufficient to restructure the encoder toward the trained objective (producing $\Delta R^2 > 0$), but the lower 3 layers are required for the encoder to recover pre-trained task quality. The additional ΔR^2 gain from $N=3$ to $N=6$ ($+0.48 \rightarrow +0.63$) is modest and has much higher variance ($0.115 \rightarrow 0.378$), consistent with the lower layers contributing primarily to task recovery rather than to the task-aligned decodability shift.

This finding sharpens the central claim: the frozen-encoder control (Appendix L) rules out data-distribution confound; the $N=3$ result rules out a whole-encoder-restructuring account.

Layer-wise drift pattern. The $N=3$ versus $N=6$ contrast reveals a within-encoder dissociation: the *top 3 layers* suffice to reorient encoder geometry toward the trained objective ($\Delta R^2 > 0$, 10/10 seeds, CKA ≈ 0.66), but the *lower 3 layers* are required for task-performance recovery ($N=3$: forg. -6.5% , 9/10 negative; $N=6$: forg. -6.7%). Both $N=3$ and $N=6$ operate in the same deeply-negative- R^2 regime; the $\Delta R^2 > 0$ signal at $N=3$ is a geometric reorientation, not a decodability gain, consistent with the main-text framing of frozen-probe ΔR^2 as directional evidence, not absolute evidence. (Probe-design robustness of Appendix L applies to both.)

Per-layer CKA breakdown. We computed per-layer CKA for condition B ($N=6$ full unfreeze, $n=10k$, $h=96$, 10 CUDA seeds). The bottom 3 layers are expected to drift more (lower CKA) than the top 3 under either the specialization or gradient-flow account; the data cannot distinguish these explanations without gradient-norm measurements.

Layer	CKA (mean $\pm$ std)	Group
1 (bottom)	0.441 $\pm$ 0.118	bottom 3
2	0.473 $\pm$ 0.109	bottom 3
3	0.514 $\pm$ 0.097	bottom 3
4	0.581 $\pm$ 0.088	top 3
5	0.647 $\pm$ 0.076	top 3
6 (top)	0.718 $\pm$ 0.064	top 3
Bottom 3 mean	0.476	
Top 3 mean	0.649	

Bottom layers drift more (mean CKA=0.476 vs. 0.649 for top 3), *despite* the top 3 being sufficient for $\Delta R^2 > 0$ ($N=3$ analysis above, 10/10 seeds). The more parsimonious explanation is standard gradient-magnitude patterns: under full backpropagation, earlier layers receive larger weight updates, producing lower CKA mechanically. Functional specialization remains an alternative hypothesis requiring a gradient-norm-per-layer comparison to distinguish from the gradient-flow null. The global CKA=0.518 at $N=6$ (Table 15) is the aggregate of these per-layer values.

O Chronos-T5-Small on M4-Monthly: Full Diagnostic Results

This appendix provides the complete results for the cross-backbone validation on Chronos-T5-Small (65M parameters, T5 seq2seq architecture with tokenised cross-entropy loss) fine-tuned on M4-Monthly (200 series, lookback=96, horizon=18). The gate-pass is 84.5% improvement over Linear baseline.

Protocol. 10 CUDA-deterministic seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), batch size 32, lr=1e-5, AdamW with weight decay 0.01, gradient clipping at 1.0, 20 epochs with early stopping (patience 5). Evaluation: batched median prediction (20 samples) on 200 held-out test windows.

Condition B (full fine-tune), $n=500$ (10 seeds).

	Forgetting%	CKA	ΔR^2	R^2 (PT)
Mean $\pm$ std	+1.6 $\pm$ 3.9	0.999 $\pm$ 0.001	-0.001 $\pm$ 0.002	+0.068 $\pm$ 0.025
Seeds neg/pos	4/10 neg forg	—	3/10 pos	10/10 pos

At $n=500$ the encoder is essentially unchanged (CKA $\approx$ 1). The positive R^2 (PT) confirms that Chronos’s pre-trained T5 encoder representations carry decodable forecasting structure, providing a second positive-floor anchor distinct from Moirai.

Condition B (full fine-tune), $n=10k$ (10 seeds).

	Forgetting%	CKA (ES)	ΔR^2	R^2 (PT)
Mean $\pm$ std	+0.5 $\pm$ 2.7	0.983 $\pm$ 0.021	$\approx$ 0	+0.047 $\pm$ 0.016
Seeds positive	6/10 forg	—	1/10	10/10 pos

At $n=10k$, early-stopped CKA averages 0.983 (vs. 0.999 at $n=500$). Only 4/10 seeds improve over zero-shot (best epoch >0 , those 4 show CKA $\approx$ 0.95–0.97); the remaining 6/10 revert to pre-trained weights under early stopping (best_epoch=0, CKA=1.000 by definition). The strong gate-pass (84.5%) means Chronos’s encoder is already near-optimal for M4-Monthly, leaving minimal room for fine-tuning improvement. While 6/10 seeds show positive (worse) early-stopped forgetting, the magnitude (+0.5 $\pm$ 2.7%) is within the noise floor established by the frozen-encoder control (condition D: +0.6 $\pm$ 5.3%); a sign test is not statistically distinguishable from chance ($p=0.75$, two-sided binomial), confirming the 6/10 pattern reflects seed-level variation rather than systematic degradation.

Final-epoch comparison (protocol robustness). Training runs to epoch 20 regardless of early stopping; we report CKA at the final training epoch to address whether encoder stability is intrinsic or an artefact of early-stopping selection:

	Final-epoch CKA	Final-epoch Forg.%	n
$n=500$ (10 seeds)	0.968 ± 0.009	$+3.1\pm 5.8$	500
$n=10k$ (10 seeds)	0.955 ± 0.006	$+3.4\pm 5.2$	10k

All 10 seeds at $n=10k$ show final-epoch $CKA > 0.94$ (min=0.945, max=0.964), confirming that encoder stability is *intrinsic* rather than an artefact of early-stopping selection. The mild CKA decline from 0.999 (early-stopped) to 0.955 (final-epoch) represents continued gradient pressure over 20 epochs, but remains far above the restructuring threshold observed on Moirai ($CKA=0.52$ on Small, 0.25 on Base). Final-epoch forgetting is $+3.4\pm 5.2\%$, mild degradation vs. the near-zero early-stopped result, but qualitatively different from Moirai-Small’s $+14\%$ at $n=500$.

Mild CKA decline interpretation. The $0.999 \rightarrow 0.955$ CKA trajectory (early-stopped $\rightarrow$ final-epoch at $n=10k$) confirms a small but real gradient signal from the fine-tuning objective. We do *not* interpret this as “no drift”; rather, the drift is quantitatively minor ($CKA > 0.94$) and functionally benign (final-epoch forgetting $+3.4\%$ vs. Moirai-Small’s $+14\%$ at $n=500$). The value-gate is consistent with this: when pre-trained features are already near-optimal (84.5% advantage), the fine-tuning gradient has little room to improve and limited capacity to harm.

Condition D (frozen encoder), $n=10k$ (10 seeds). Forgetting: $+0.6\pm 5.3\%$ (5/10 negative). $CKA=1.000$ (encoder frozen by definition), $\Delta R^2=0$. Decoder-only adaptation is unreliable: neither consistently better nor worse than zero-shot across seeds.

Random-init control (6 seeds). $CKA=0.000$ (expected: random weights have no similarity to pre-trained). $R^2(PT) < 0$ (-0.004 ± 0.005), confirming that positive-floor R^2 on the pre-trained encoder requires the learned representations, not the architecture alone. Training loss remains high (> 8.0) and predictions are incoherent, confirming that Chronos’s pre-trained representations are load-bearing.

Comparison with Moirai. The contrast validates the value-gate as a regime indicator: *Moderate gate-pass* (Moirai/ETTh2, 28%): encoder restructures aggressively ($CKA=0.52$), probe asymmetry emerges ($\Delta R^2 = +0.67$, orthogonal < 0), forgetting is capacity- and protocol-dependent. *Strong gate-pass* (Chronos/M4, 84.5%): encoder is stable ($CKA \geq 0.95$ even at final epoch), no measurable probe shift, forgetting near-zero under early stopping and mild ($+3.4\%$) at final epoch; fine-tuning is unnecessary but not harmful. Both outcomes are correctly predicted by the gate criterion, demonstrating that the diagnostic toolkit generalises across architecturally distinct backbones. The probe asymmetry mechanism (task-specific $\Delta R^2 > 0$) requires encoder restructuring to produce a measurable signal; on Chronos, encoder stability means the asymmetry is not observable; this is a correct null prediction, not a failure of the diagnostic.

O.1 ILI Gate Check

Moirai-Small zero-shot on ILI (national illness, 966 weekly observations, 7 features, $h=24$, lookback=104) yields gate-pass=57% (ZS MSE=1.183, Linear MSE=2.748, per-feature normalisation by context statistics).

Full probe results (10 CUDA seeds, conditions B/D). Table 16 reports per-seed aggregate diagnostics. All 10 seeds show $\Delta R^2 > 0$ (aggregate: $+0.123\pm 0.046$; Wilson CI [0.72, 1.00]) and forgetting < 0 ($-80.7\pm 0.9\%$; 10/10 negative, meaning fine-tuning dramatically improves on ZS). $CKA=0.478\pm 0.022$ places ILI in the restructuring regime, not the predicted intermediate regime (0.90–0.95). Condition D (frozen encoder): $CKA=1.000$, $\Delta R^2=0$, forgetting=0% in all 10 seeds, confirming encoder weight updates drive all changes.

Per-feature breakdown. OT (target feature) shows the largest ΔR^2 ($+0.73\pm 0.16$, 10/10 positive); AGE 0–4 also shows consistent gain ($+0.15\pm 0.05$, 10/10). AGE 5–24 shows slight negative signal (-0.07 ± 0.09 , 3/10 positive), consistent with the task-specificity interpretation: features less aligned with the forecasting objective show reduced or negative decodability shift.

Table 16: ILI per-seed results (condition B, Moirai-Small, $h=24$, lookback=104). Aggregate ΔR^2 is the mean across 7 univariate features.

Seed	CKA	ΔR^2	Pos. features	Forg.%	Drift
42	0.465	+0.211	6/7	-81.5	5.61
101	0.465	+0.123	3/7	-80.9	5.34
123	0.511	+0.087	3/7	-82.3	5.13
202	0.461	+0.169	5/7	-79.6	5.67
303	0.498	+0.169	7/7	-80.9	5.49
456	0.454	+0.066	5/7	-80.2	5.63
777	0.498	+0.074	4/7	-80.1	5.36
789	0.460	+0.144	6/7	-79.8	5.49
888	0.459	+0.111	4/7	-79.7	5.24
999	0.512	+0.080	4/7	-82.0	5.25
Mean $\pm$ std	0.478 $\pm$.022	+0.123 $\pm$.046	4.7/7	-80.7 $\pm$ 0.9	5.42 $\pm$.21

Interpretation. The pre-registered prediction had two components, and neither now supports the hypothesis. The CKA component (0.90–0.95, intermediate behaviour) is **falsified**: CKA is in the restructuring regime (0.478), not intermediate. The probe-asymmetry component ($\geq 7/10$ seeds $\Delta R^2 > 0$) was reported as confirmed, but rests on the statistic we retract (§6) and is therefore **with-drawn** rather than confirmed. The prediction as a whole is falsified, not partially so. This revises the gate-pass spectrum: the restructuring-to-stability transition occurs above $\approx 80\%$ gate-pass, not at 50–70%. ILI’s low CKA reflects the small dataset (966 timesteps, ≈ 450 training windows per feature) which permits aggressive restructuring within 20 epochs.